CertJudge: Evaluating Lean Formal-Code With Falsifiable Properties

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Validating LLM judges one human study at a time does not scale

Agents write Lean 4 code and its specs faster than humans can inspect — can one human calibration, via automatic falsifiable checks, be amortized across every future judge?

Key idea: certify the judge — don't re-label it

- **CertJudge**: rate a small panel **once**.
- **Automatic falsifiable property checks** + $\mathrm{TI}_{\mathrm{var}}$ certify every new judge — no new labels.
- Calibration: $n{=}75$ candidate–gold Lean pairs, single annotator, two rubric passes; 8 judge configs per family panel.

Four falsifiable properties in a trust index

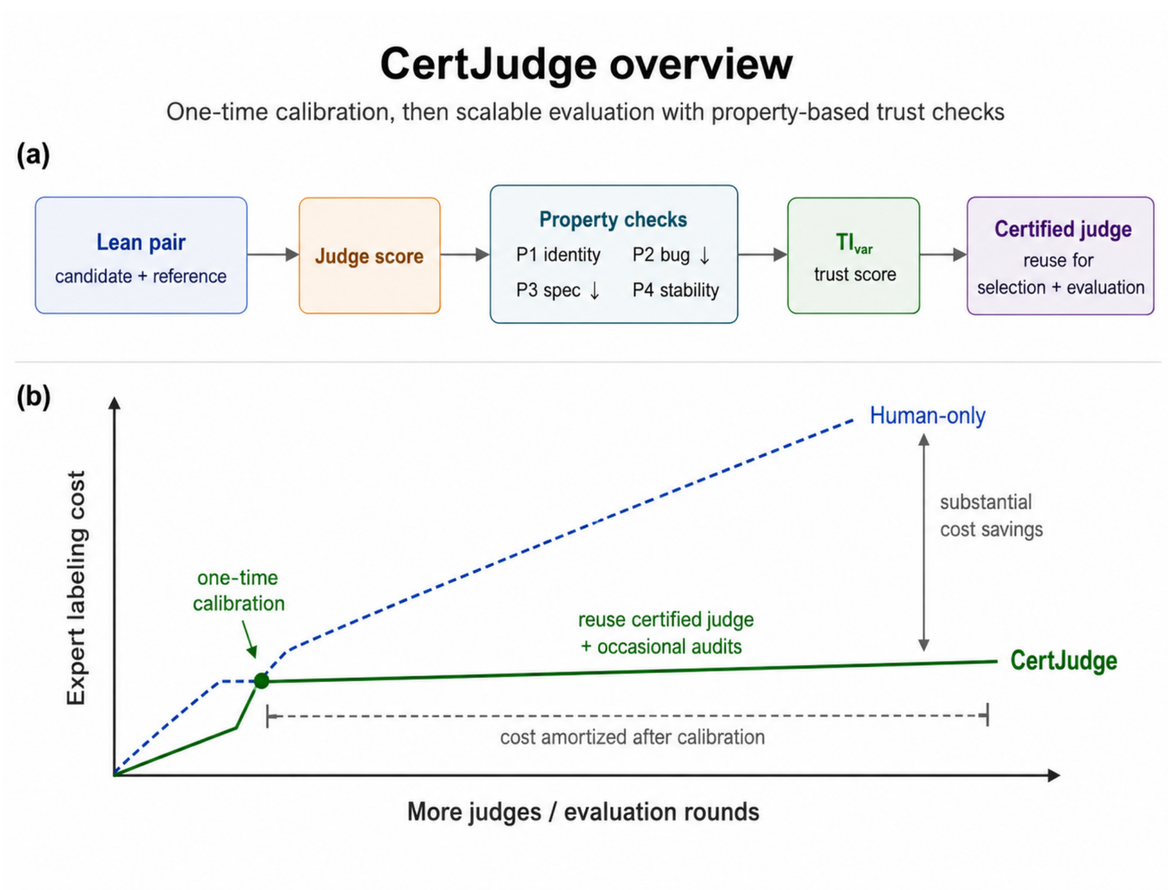


Figure 1. Label once, then certify automatically — labeling cost stops growing with the number of judges.

P1 Identity: $s(X, X) = 1$

P2 Bug monotonicity: score ↓ as bugs injected

P3 Spec monotonicity: score ↓ as specs removed

P4 Stability: invariant across repeats & rewrites

$$\mathrm{TI}_{\mathrm{var}} = \exp\Big(\tfrac{1}{3}\log C_f + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarCorr}}) + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarScore}})\Big)$$

- C_f aggregates P1–P4 into a single geometric-mean score.
- **One failed property collapses the index.**

Result: $\mathrm{TI}_{\mathrm{var}}$ predicts judge-level human alignment

$$\rho = 0.833$$

validation-set Spearman between $\mathrm{TI}_{\mathrm{var}}$ and human agreement
avg labels; 0.905 pass1 / 0.738 pass2

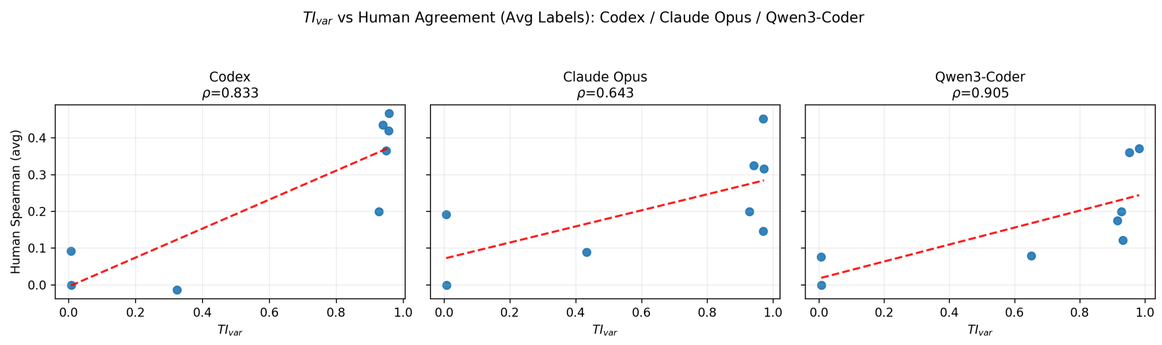


Figure 2. Higher $\mathrm{TI}_{\mathrm{var}}$ → higher human agreement in **all three judge families**.

Catches what agreement misses: a broken judge hits ≈ 0.45 human Spearman yet fails weak monotonicity (12.5%).

Stress tests keep the judge honest

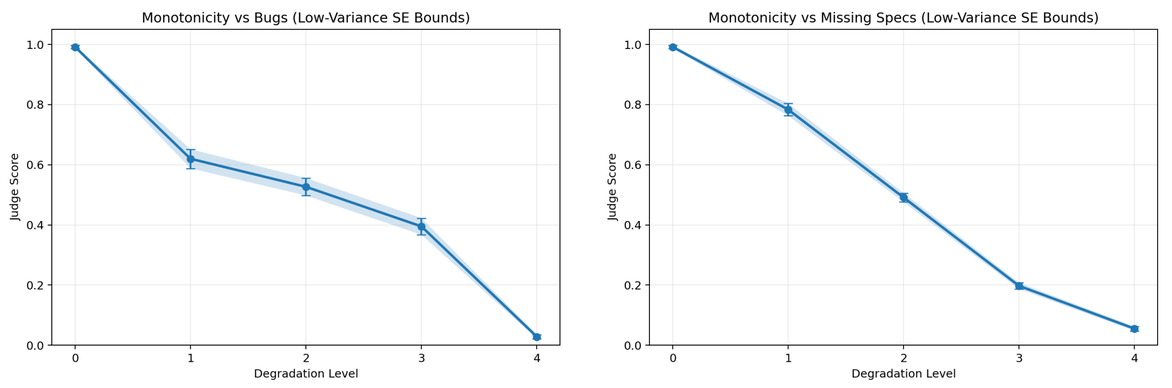


Figure 3. GPT-5.3-Codex judge, 184 VeriBench files [1]: scores fall as bugs are injected ($\rho = -0.808$) and as specs are deleted ($\rho = -0.899$).

Known failure mode: aggressive α-renaming scores 0.399 vs. ≥ 0.849 for milder rewrites.

Certified judge ranks agents: **DSPy ReAct 0.615** > Trace++ 0.588 > Trace+ 0.477 > **baseline 0.470**.

Takeaways

- **Amortized oversight:** one $n{=}75$ panel + automatic falsifiable checks screen every new judge — no re-labeling.
- **Measure trust, don't assume it:** behavioral properties expose failures human-agreement scores miss.
- **Scope:** *validation-set calibration evidence, not universal reliability* — locked held-out test (5 raters, 343 labels, 78 pairs) is next.

References

[1] Brando Miranda, Srivatsava Daruru, Ethan S Hersch, Zhanke Zhou, Allen Nie, Daneshvar Amrollahi, Leni Aniva, Iddah Mlauzi, Kirill Acharya, Elyas Obbad, et al. VeriBench: An end-to-end formal verification benchmark for AI coding agents in Lean 4. In *3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents*.

[2] Weixi Tong and Tianyi Zhang. CodeJudge: Evaluating code generation with large language models, 2024.

[3] Lianmin Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, 2023.

Code